Some GIS related Terminology

- **ArcGIS**: *[software]* ArcGIS is a GIS software package produced by the Environmental Systems Research Institute (Esri) which allows us to collect, store, manage, visualize, export, analyze and map geographic data. ArcGIS is a suite of spatial analysis tools and cartographic features on a customizable user interface that allows us to be understand and make decisions with spatial data.

- **ArcCatalog**: *[software]* ArcCatalog is an application in the ArcGIS suite used to manage geographic data – similar to windows file explorer.
- **ArcGlobe**: *[software]* ArcGlobe is a global three-dimensional visualization and analysis environment as part of the Esri ArcGIS suite (3D analyst), specializing in global datasets and larger study areas.
- **ArcScene**: *[software]* ArcScene is a 3D feature and raster viewer part of the Esri ArcGIS suite of applications (3D analyst) specializing in small study area scenes.

- **Aspect**: *[GIS processing]* Aspect is the slope direction on a terrain surface. Aspect is measured clockwise starting North as 0° to 360° North again with flat areas given a value of -1 (or 0 degrees).
- **Attribute table**: *[data structure]* An attribute table stores non-spatial information in columns and rows about geographic data – similar to spreadsheets.

- **Basemap**: *[cartography]* A basemap is a background georeferenced image that gives a point of reference on a map. Basemaps are non-editable and provides aesthetic appeal such as aerial imagery, topography, terrain and street layers.

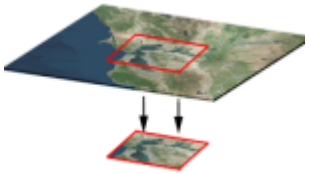
- **Boolean Algebra**: *[mathematics]* Boolean algebra are conditions used to select features with a set of algebraic conditions. Conditions include AND, OR or NOT. Venn diagrams are often used to represent Boolean operations.

- **Buffer**: *[GIS processing]* The buffer tool is a proximity function that creates a polygon at a set distance surrounding a selected feature or features.

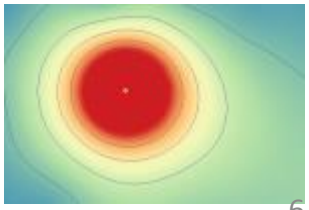
- **Cartography**: *[GIS dictionary]* Cartography is the study and science of representing real-world entities on maps. Cartography combines aesthetics and science to communicate spatial information in two or three dimensions.

- **Centroid**: *[geometry]* A centroid (or geometric center) is a central point of an area feature. Centroids are defined as the average position of all the points in the shape.

- **Clip**: *[geometry]* A clip is an overlay tool that involves clipping an input layer with the extent of a defined feature boundary. The result of this tool is a new clipped output layer.



- **Contour Line**: *[data structure]* A contour line is a constant value for mapping any variable such as elevation and temperature maps. Contour lines are used to map equal values and illustrate topography or relief on a map. They are also known as isopleth or isoline maps.

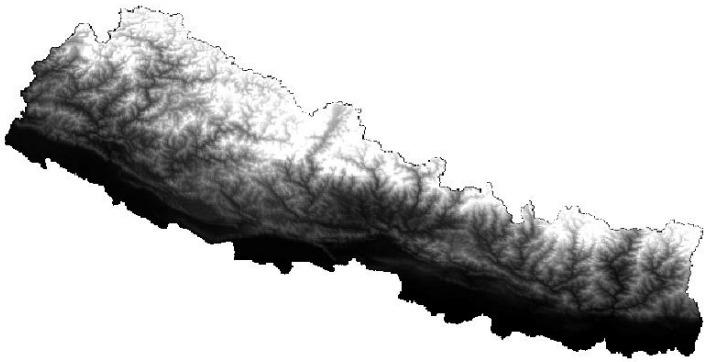


- **Control point**: *[coordinate system]* Control points are XY locations known to have a high degree of accuracy. They are used to convert digitized coordinates from paper maps and georeferencing to standard map projection coordinates.
- **Coordinates**: *[coordinate system]* Coordinates are pairs (X, Y) or triplets (X, Y, Z) of values that are used to represent points and features on a two and three-dimensional space. The X-value represents the horizontal position and Y-value represents the vertical position. The Z-value generally refers to the elevation at that point location.

- **Data Model**: *[data structure]* The two main GIS data models are rasters (grids) and vectors. Rasters are sets of pixels with a specific cell size. Vectors represent points, lines and polygons.

- **Database Management System (DBMS)**: *[data structure]* A DBMS is a collection of tools that permits the entry, storage, input, output and organization of data. It serves as an interface between users and their database.

- **Digital Elevation Model (DEM)**: *[GIS processing]* A DEM is a bare earth elevation model representing the surface of the Earth. DEMs filter out non-ground points such as bridges and trees.

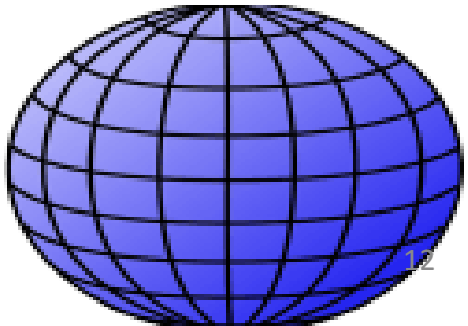


- **Digital Surface Model (DSM)**: *[file format]* A digital surface model is an elevation that includes the top of buildings, tree canopy, powerlines and other features above the bare earth. For example, the first return of LiDAR consists of a DSM.

.

- **Digitize**: *[editing]* When digitize in a GIS, youare creating geographic computer-compatible stored data with lines, points and polygons in a spatial database.
- **Easting**: *[coordinate system]* Eastings refer to the x-axis (eastward) approximately parallel to lines of equal latitude commonly used in the Universal Transverse Mercator map projection. A pair of coordinates eastings and northings refers to a Geographic coordinate point on Earth.

- **Ellipsoid**: *[surveying]* Reference ellipsoids are mathematical models of the shape of the Earth with the major axis along the equatorial radius. They are primarily used as a surface to specify point coordinates such as latitudes (north/south), longitudes (east/west) and elevations (height).
- **Entity**: *[cartography]* Entities represent real-world point, line or polygon features with a geographic location. Entities are stored in databases, where they can be retrieved and displayed using GIS software.



- **False northing**: *[coordinate system]* A false northing is a number added in a map projection in the y direction to avoid negative coordinate locations within the specified area of that map projection.
- **Feature**: *[data structure]* A feature is a cartographic point, line or polygon object with a spatial location in the real-world landscape that can be used in a GIS for storage, visualization and analysis.

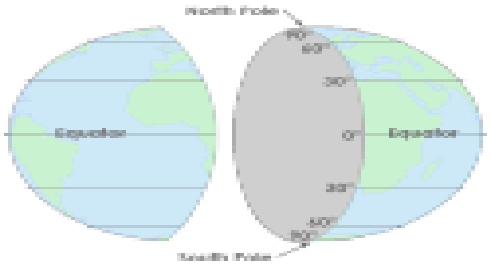
- **Geocentric**: *[geodesy]* Geocentric is a measuring system that uses the center of the Earth as its origin. The WGS 1984 datum is a true geocentric datum while the NAD83 datum is offset by about two meters from the center of the Earth.

- **Geocoding**: *[GIS processing]* Geocoding is the process of assigning geographic coordinates to places based on street address, town/city, province/state and country.

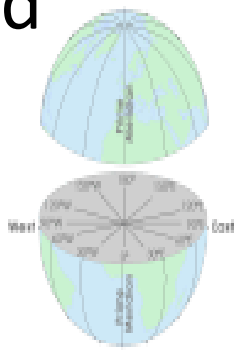
- **Graticule**: *[coordinate system]* Graticules are lines of latitude or longitude on a digital or hard copy map and assist in showing the geographic locations of map features.

- **Greenwich Meridian (Prime Meridian)**: *[coordinate system]* The Greenwich meridian is a line of longitude that passes through the Royal Observatory in Greenwich, England. In a geographical coordinate system, it is a line of longitude defined to be 0°.

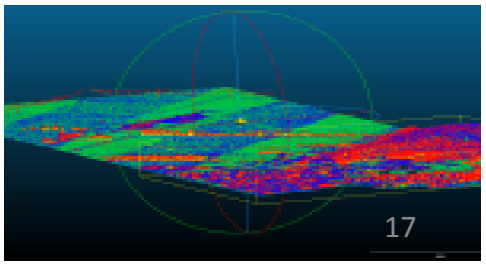
- **Interpolation**: *[mathematics]* Interpolation is the estimation of unsampled locations based on known location sampled values. Interpolation is often used in deriving elevation, temperature and other predicted variables in a grid.
- **Latitude**: *[coordinate system]* Latitudes are spherical coordinates of Earth locations that vary in North-South directions. Lines of latitude are angles on the Earth's surface which ranges from 0° at the Equator to 90° (North or South) at the poles.

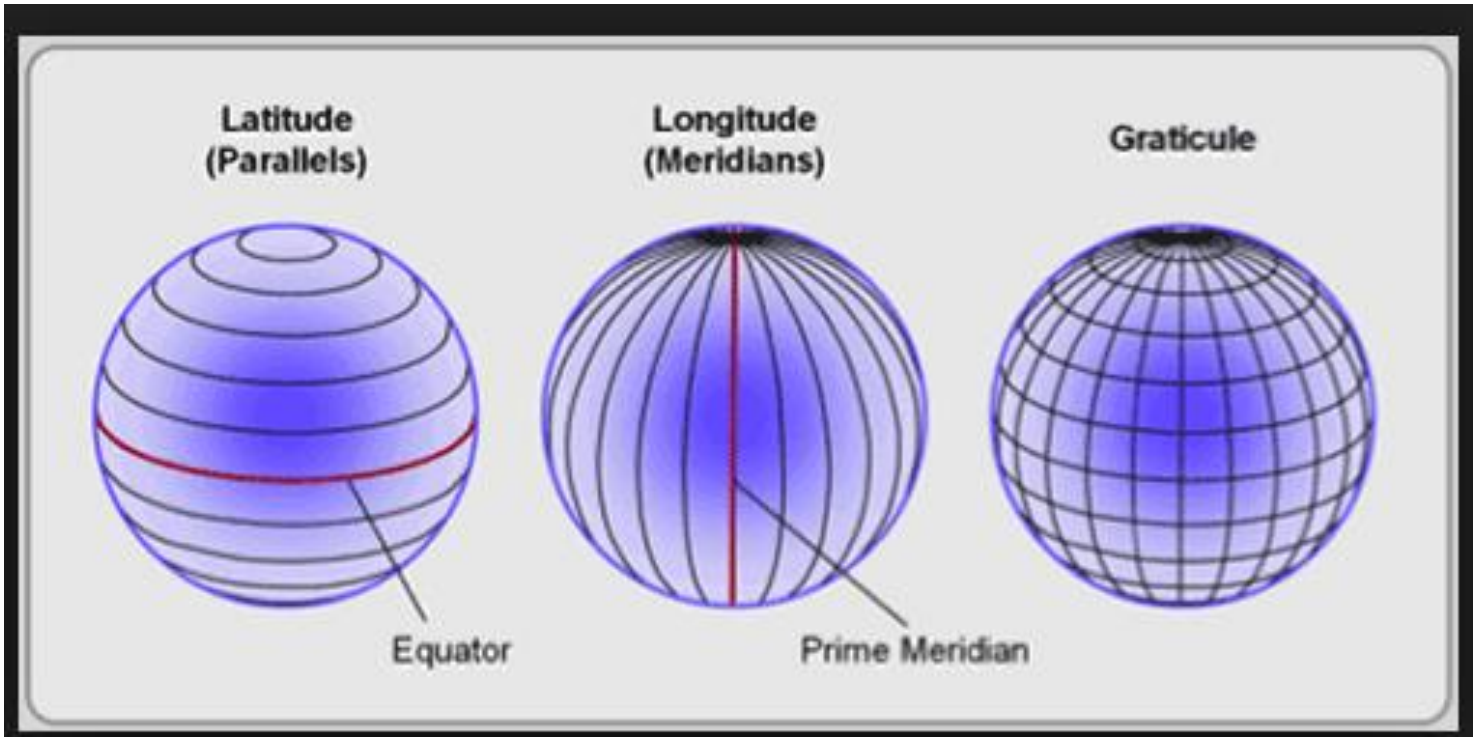


- **Longitude**: *[coordinate system]* Longitudes are coordinate on Earth that vary in east-west directions. They are usually expressed in degrees ranging from 0° to +180° east and −180° west. The prime meridian marks the 0° longitude and passes through the Royal Observatory in Greenwich, England.



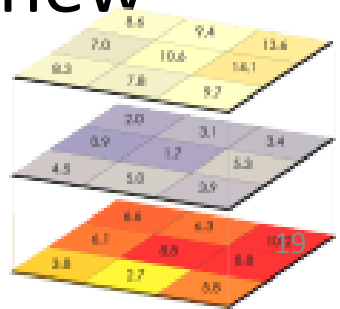
- **Light Detection and Ranging (LiDAR)**: *[remote sensing]* LiDAR uses laser pulse measurements to identify heights, depths and other properties of features on the Earth's surface. Example outputs for LiDAR are digital elevation models, light intensity and digital surface models.



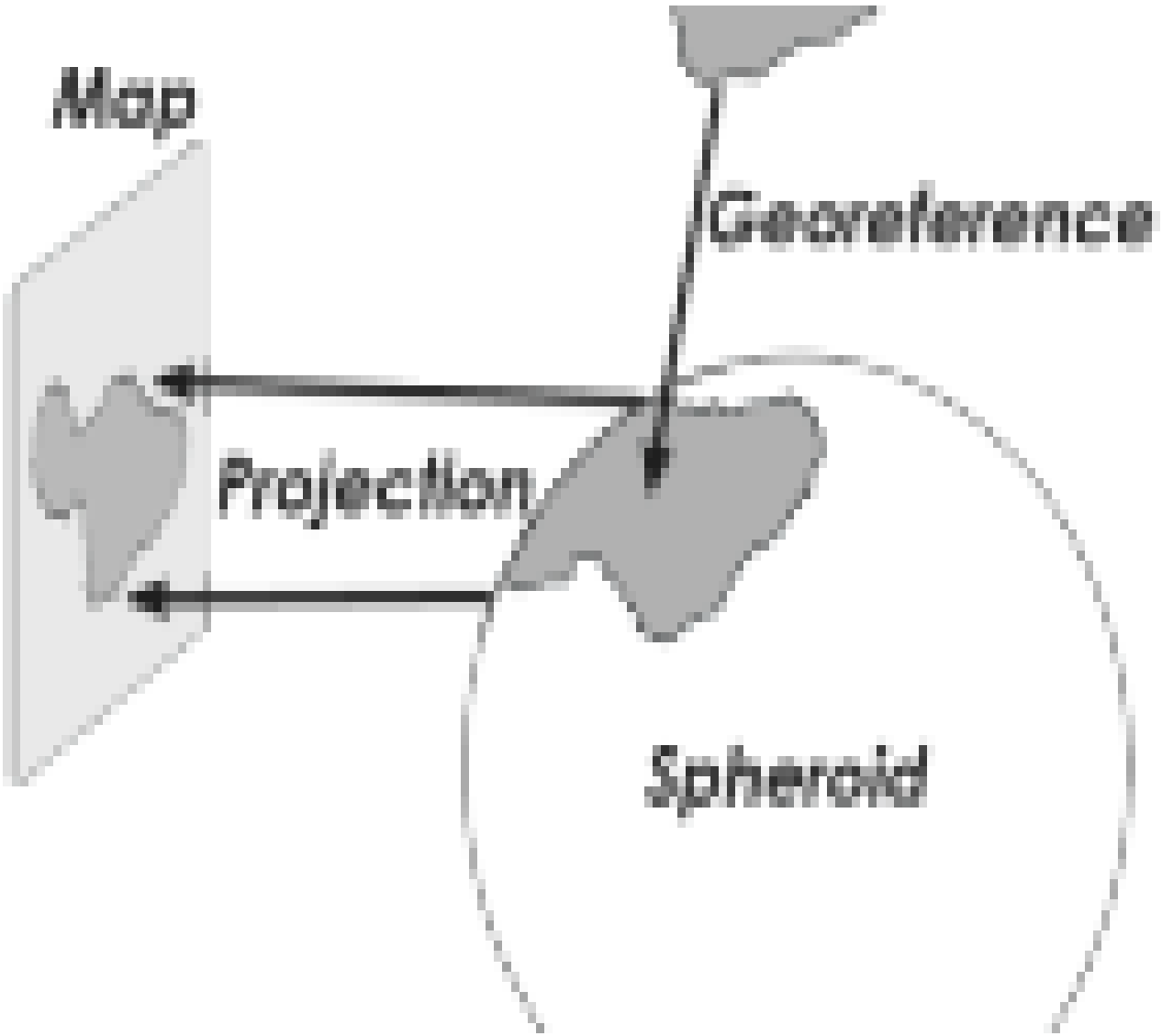

Latitude
(Parallels)
Longitude
(Meridians)
Graticule
Equator
Prime Meridian

- **Locator Map**: *[cartography]* Locator maps show the extent of the primary map in relation to a larger extent area. It provides context to the map reader to gain a better understand of the overall area.

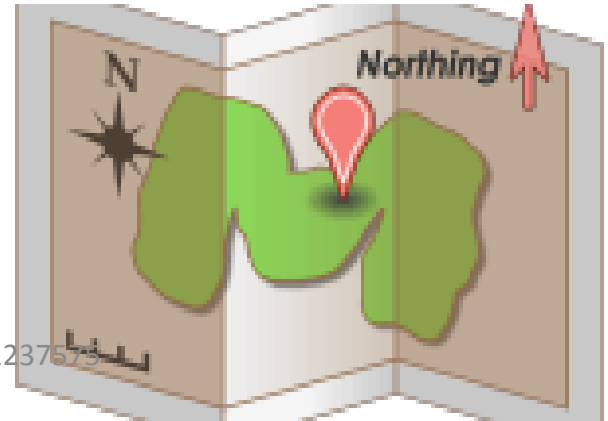
- **Map Algebra**: *[mathematics]* Map algebra is the combination of spatial data using spatial and mathematical, statistical and trigonometric operations to generate new raster outputs.



- **Map Legend**: *[cartography]* A map legend is a visual graphic of the symbology used in a map that tells the map reader what the polygons, lines, points or grid cells represents.
- **Map Projection**: *[cartography]* A map projection is a systematic rendering of features that renders a 3D ellipsoid or spheroid of Earth to a 2D map surface. Because 3D surfaces cannot be displayed perfectly in a two-dimensional space, some distortions of conformality, distance, direction, scale, and area always results from map projections.


Map
Georeference
Projection
Spheroid

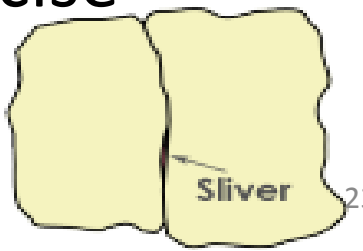
- **Metadata**: *[data structure]* Metadata is data about data. Metadata describes the characteristics of a dataset which includes abstract, coordinate system, attribute information, origin and accuracy.
- **Northing**: *[coordinate system]* A northing is the axis in the approximate north-south direction (or y-value) in UTM and other standard coordinate systems.



- **Overshoot**: *[editing]* An overshoot is a digitized line that extend past the intended line of connection.

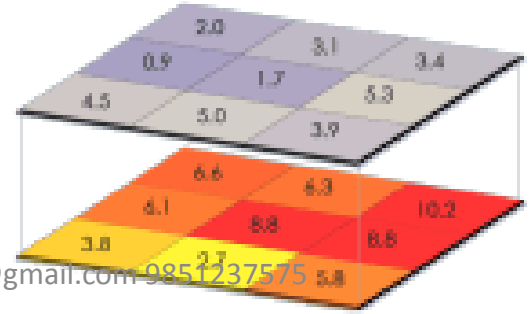
- **Undershoot**: *[editing]* An undershoot is considered to be a digitizing error in which the new line falls short of the intended connection segment.

- **Sliver**: *[editing]* A sliver is a small, spurious gap considered as a topology error. Slivers are found between polygons as a result of imprecise digitization of features in GIS.



- **Panchromatic Image**: *[editing]* A panchromatic image records wavelengths in only one band resulting in grayscale images.

- **Pixels (GIS)**: *[data structure]* Pixels are the grid cells that make up raster images. Each cell is identical in size and records a brightness and color in an image.



- **Query**: *[programming]* A query is a request or search of spatial or tabular data based on user-define criteria. For example, the results of a query is a subset or selected records. SQL, Python and Java are programming languages how queries are made.
- **Registration** : *[GIS processing]* Registration is the alignment or assignment of coordinates from a non-projected coordinate system to a coordinate system.

- **Selection Tool**: *[GIS processing]* The selection tool identifies an object or set of objects based on user-selected criteria and properties, typically with Structured Query Language (SQL).

- **Slope**: *[mathematics]* Slope is the change in elevation with respect to change in location measured in degrees or percent slope. Slope indicates the incline/decline and steepness of the surface, typically measure in a fixed interval.

- **Spatial Relationship**: *[data structure]* A spatial relationship links GIS features with a table by a unique identifier all stored in a relational database management system.
- **Structured Query Language (SQL)**: *[map projection]* SQL is a set of commands used to manage data stored in relational database management systems (RDBMS).
- **Thematic Layer**: *[data structure]* A thematic layer is a distinct spatial entity in a data layer. They are usually delineated as points, lines and polygons.

- **Topology**: *[editing]* Topology is a set rules that dictates the spatial properties of point, line and area features such as connectivity, adjacency and contiguity.

- **Universal Transverse Mercator (UTM)**: *[coordinate system]*UTM coordinate system is a standard set of map projections with a central meridian for each six-degree wide UTM zone.